

The Profile Bulging PyRoLL Plugin

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This plugin predicts the free-surface ("bulge" / rounding) shape of a profile's cross-section after a three-roll (Kocks mill / flat-roll block) rolling pass, i.e. the part of the boundary that is not in contact with the rolls.

1 Model scope

All models implemented here predict the free-surface shape *given* the maximum width B_1 of the profile after rolling. How B_1 itself is predicted (spread prediction) is a separate concern, handled by other PyRoLL plugins such as `pyroll-wusatowski-spreading`; this plugin only reshapes the free surface of the cross-section consistently with a given (already solved) width. This mirrors how the source papers themselves separate the two problems: a spread formula calibrated against FE data, and a free-surface-profile formula that consumes the predicted spread as an input.

A generic spread formula such as Wusatowski's needs the equivalent height and width of the incoming and outgoing profile, which `pyroll-core` does not define for arbitrary (in particular 3-fold symmetric) cross-sections on its own; `pyroll-lendl-equivalent-method` supplies them via Lendl's equivalent rectangle method, including a `ThreeRollPass`-specific formula. Loading it together with a spread-prediction plugin lets a `ThreeRollPass` predict its own out-profile width instead of requiring it to be set explicitly, and is recommended for realistic use; Section 3 demonstrates the full chain (Lendl + Wusatowski + this plugin) end to end.

2 Three-roll passes

Three-roll (Kocks) mills roll a workpiece between three rolls arranged 120° apart about the pass line, closing in on the stock symmetrically. Two variants are modeled, following Min, Kwon, Lee, Woo, and Im [1] and Byon, Kim, Kim, and Lee [2] respectively:

- **Flat-roll mills** [1]: the roll surface is flat along its axis (no machined groove); a round bar is progressively formed into a hexagonal shape over several passes. `pyroll-core`'s `FlatGroove` used in a `ThreeRollPass` models this case ("`flat`" classifier).

- **Kocks mills with a curved (grooved) roll surface** [2]: the roll surface carries a shallow, machined groove (radius $D_g/2$), producing a rounded ("Koval") caliber. Any non-flat groove used in a **ThreeRollPass** (round, oval, hexagonal, ...) is treated this way.

Both papers distinguish three types of geometry transition, based on the shape of the incoming profile and of the current pass:

1. **round** $\rightarrow$ **curved-hexagonal** / **Koval** – the incoming profile is round ("round" in its classifiers).
2. **curved-hexagonal** $\rightarrow$ **hexagonal** / **Koval** $\rightarrow$ **Koval** – neither the incoming profile nor the current pass is round.
3. **Koval** $\rightarrow$ **round** – the current pass itself is round (the final pass of a Kocks mill schedule). Min et al. do not treat this case, since their experiments never return to a round shape.

2.1 Flat-roll mills (Min et al. 2003)

For the round $\rightarrow$ curved-hexagonal pass, the free surface is modeled as a circular arc, whose center of curvature is offset from the profile's center by an eccentricity e :

$$\frac{e}{B_0} = 2.40 \cdot \frac{F_H}{F_O}, \quad R_1 = B_1 - e,$$

where B_0 is the incoming profile's own radius (half its maximum width), B_1 the radius of the outgoing profile after rolling, F_O the incoming profile's cross-sectional area and F_H the part of it that does not fit the new pass's usable cross-section (`roll_pass.displaced_cross_section` in `pyroll-core`), i.e. the area the roll must displace.

For the curved-hexagonal $\rightarrow$ hexagonal and hexagonal $\rightarrow$ hexagonal passes, Min et al. approximate the free surface as a *straight chord* instead, from a point W_1 on the roll face (where contact ends) to the corner tip B_1 . This plugin locates W_1 as the point where the roll face crosses the circle of radius B_1 around the profile's center, on each of the two roll faces flanking a given corner, and builds the polygon directly from the resulting W_1, B_1, W_1 vertex triples at each of the three corners; between corners the boundary follows the (straight) roll face exactly, requiring no further approximation there. Min et al. note this linear approximation slightly under-predicts the true (mildly curved) bulging in this case, with less than 2% error in cross-sectional area.

2.2 Kocks mills (Byon et al. 2017)

All three transition types are modeled as a circular arc with radius of curvature R_s and center offset P_c from the profile's center:

$$\text{round} \rightarrow \text{Koval} \quad \frac{P_c}{B_0} = 3.133 \cdot \frac{F_H}{F_0}, \quad R_s = B_1 - P_c.$$

Koval $\rightarrow$ **Koval** $P_c = 0$, $R_s = B_1$ (a circle through the section's own center).

Koval $\rightarrow$ **round** P_c and R_s solve the system

$$R_s = \frac{DS}{2} + P_c, \quad (B_1 \sin 60^\circ)^2 + (B_1 \cos 60^\circ + P_c)^2 = R_s^2,$$

where DS is the current pass's own inscribed circle diameter (`roll_pass.inscribed_circle_diameter`) giving the closed form

$$P_c = \frac{(DS/2)^2 - B_1^2}{B_1 - DS}.$$

2.3 Implementation

The dispatch between these cases (module `pyroll.profile_bulging.three_roll_pass`) is driven purely by classifiers already available on the roll pass and its incoming profile – no additional configuration is required:

Condition	Flat roll	Curved (Koval) roll
"round" in <code>in_profile.classifiers</code>	Min circular arc	Byon round $\rightarrow$ Koval
"round" in <code>roll_pass.classifiers</code>	(not treated by Min)	Byon Koval $\rightarrow$ round
otherwise	Min linear chord	Byon Koval $\rightarrow$ Koval

The resulting eccentricity and radius of curvature are exposed as the hooks `ThreeRollPass.OutProfile.b` and `ThreeRollPass.OutProfile.bulge_radius` (the latter is `None` for Min's linear-chord passes, which have no single radius of curvature).

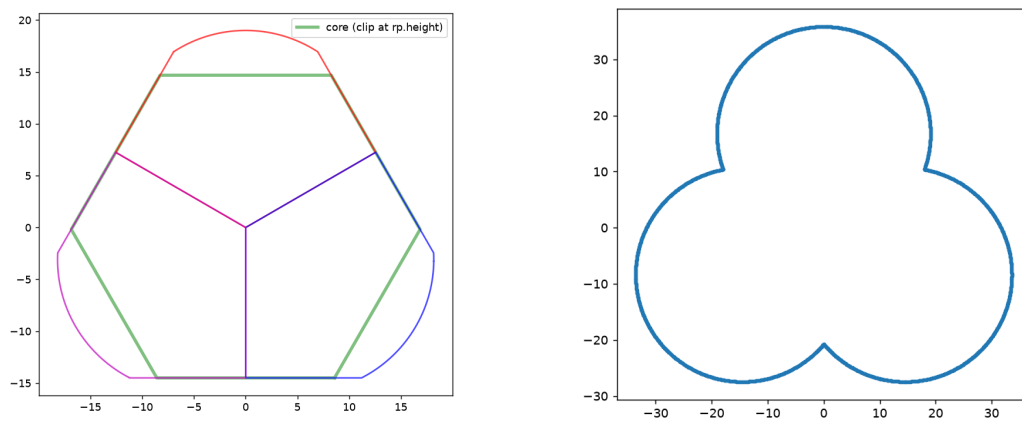
For the circular-arc cases, the free surface is traced directly, angle by angle, as whichever is closer to the profile's center: the nearest corner's bulge circle, or the roll's own unbulged contour (full roll contact). This is deliberately *not* built via the union of the three, generally overlapping, corner circles intersected with the roll contour, since that is prone to a spurious reentrant cusp wherever two such shapes cross without being exactly tangent there (see Fig. 1 and the discussion in Section 2.4). Sampling the two candidate radii directly instead guarantees a shape with no self-crossings and keeps the residual non-smoothness to a minimum, but cannot remove it entirely: two neighboring corners' bulge circles are not, in general, tangent to each other or to the roll contour at the point where the model hands off between them, so a small change in slope at that point is an inherent feature of connecting three independently-fitted per-corner arc formulas into one closed cross-section – most visible for a heavily-reduced first (round-input) pass, where the eccentricity is large relative to the spacing between corners (see the round $\rightarrow$ Koval case in Fig. 3). The source papers validate each corner's own arc against a one sixth-symmetric sector of FE/experimental data (Section 3) and do not themselves address the smoothness of the reassembled full cross-section. Section 2.4 shows, with figures, exactly how each corner's arc is delimited and handed off to its neighbor; Section 2.5 covers what happens when the predicted width makes a corner's own arc inconsistent with this construction altogether.

2.4 Construction method

Figure 1 illustrates the two building blocks used above, both from early exploratory scripts kept here for their explanatory value (they are not part of the shipped model). The left panel shows Min's linear-chamfer construction (Section 2.1) for one corner of a curved-hexagonal $\rightarrow$ hexagonal pass: the plain hexagonal roll-face contour (green, the "`core (clip at rp.height)`" in the legend) is replaced near the corner by the two straight chords from W_1 (magenta/blue, where each roll face crosses the circle of radius B_1) to the corner tip B_1 itself, exactly as implemented in `linear_chamfer_cross_section`. The right panel shows why the circular cases are *not* built by simply assigning each of the three corner circles (radius R_s , center offset by eccentricity P_c) to its own 120° sector without further restriction: for a large eccentricity relative to the corner spacing, as in a heavily reduced round $\rightarrow$ Koval first pass, each corner's own circle is only valid close to its own tip, but stays numerically defined – and numerically smaller than it should be – all the way into the neighboring corners' territory. Simply switching between corners at the sector boundary then produces exactly this kind of "clover", pinched down to a fraction of the corner radius exactly between corners, rather than the plausible, convex shape the source paper actually depicts – this is, in fact, precisely the unphysical shape a predicted width can trigger if the plugin does *not* check for it (see Section 2.5 below): restricting each corner's circle to its own corner-local range via `corner_crossing_offset`, and falling back entirely when even that corner-local range cannot be made consistent with the roll's natural contour, is what avoids ever emitting a shape like this one.

2.5 Fallback for inconsistent predicted widths

Because the predicted width B_1 comes from an independent, separately calibrated spread model (Section 1), it is not guaranteed to be consistent with the eccentricity/radius this plugin computes from it. Before tracing the free surface, `corner_crossing_offset` searches outward from a corner, up to its own 60° half-sector, for the angle at which that corner's own bulge circle first becomes at least as large as the roll's natural (un-bulged) contour; beyond that offset the plain contour is used instead (this is the mechanism restricting each corner's circle to its own corner-local range, referenced in Section 2.4 above). If no such crossing exists anywhere within the half-sector – i.e. the circle stays smaller than the natural contour all the way into the *next* corner's own territory – or if the computed radius of curvature itself comes out non-positive, the predicted width is judged inconsistent with the model for this pass, and `ThreeRollBulgingModel.solve` raises `BulgeModelNotApplicable` internally and catches it, logging a warning and falling back to returning the profile with its plain, un-bulged cross-section from `pyroll-core` unchanged, rather than emitting the "necked" shape of Fig. 1(b). In this case `ThreeRollPass.OutProfile.bulge` and `.bulge_radius` are both `None`, which calling code can check to detect that the fallback was used. `tests/test_three_roll_byon_koval.py`'s `test_byon_pass_1_round_to_koval` exercises this directly: Byon's own round $\rightarrow$ Koval eccentricity formula, applied to a heavily reduced first pass (71 mm round stock into a 59.9 mm inscribed-circle Koval groove), predicts an eccentricity so large that no crossing exists for any reasonable groove shape



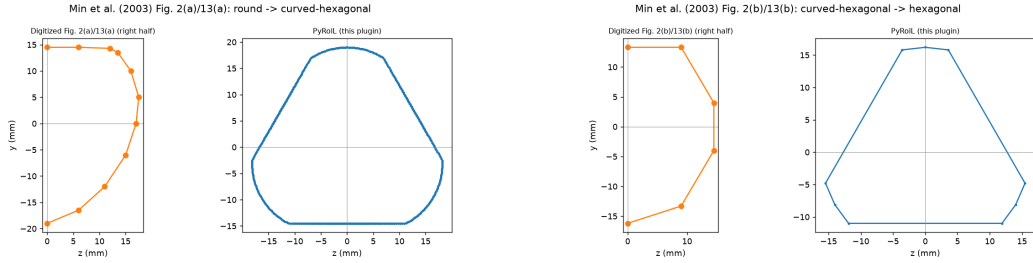
- (a) Linear-chamfer construction (Min, curved-hex $\rightarrow$ hex): the hexagonal core contour (green) is replaced near each corner by the two chords to W_1 (colored per face) and on to B_1 .
- (b) Naively assigning each of the three corner circles to its own 120° sector, unrestricted (large-eccentricity round $\rightarrow$ Koval case): a pinched "clover", exactly the kind of unphysical shape `corner_crossing_offset` and the Section 2.5 fallback exist to prevent.

Figure 1: Illustrative constructions behind the linear-chamfer and circular-arc cross-section builders (exploratory scripts, not shipped model code).

at that inscribed circle diameter; Fig. 3(a) and the overlay in Fig. 5 show the resulting fallback (angular, core) shape next to the true digitized bulge, honestly showing where the model declines to guess rather than silently producing an implausible shape.

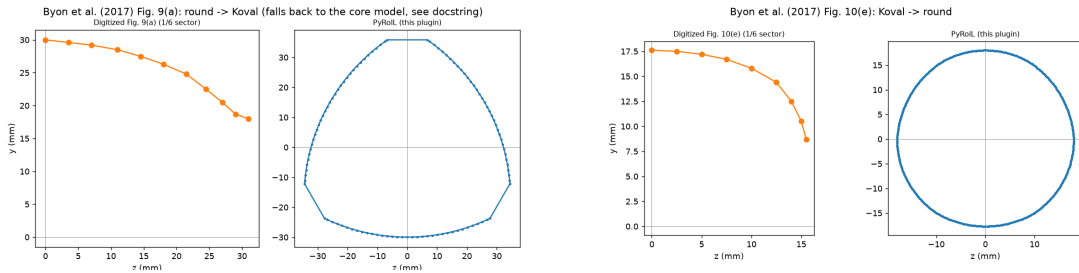
3 Validation

The three-roll models are validated against the figures of the source papers by reproducing their process parameters in PyRoLL and comparing the predicted cross-section against boundary coordinates digitized from the printed figures. The out-profile width is pinned to the digitized shape's own spread tip radius for each case (bypassing spread prediction, see Section 1), so that only the free-surface model itself is being validated. See `tests/test_three_roll_min_flat.py`, `tests/test_three_roll_byon_koval.py` and `tests/test_three_roll_koval.py` for the full test code and digitized data (`tests/data/`); running the test suite regenerates the comparison plots under `tests/output/`.



(a) Min et al., round $\rightarrow$ curved-hexagonal (circular arc). (b) Min et al., curved-hexagonal $\rightarrow$ hexagonal (linear chord).

Figure 2: PyRoLL prediction (right) vs. digitized Fig. 2/13 of Min, Kwon, Lee, Woo, and Im [1] (left).



(a) Byon et al., round $\rightarrow$ Koval (pass 1) – falls back to the plain core cross-section, see Section 2.5. (b) Byon et al., Koval $\rightarrow$ round (pass 7, final).

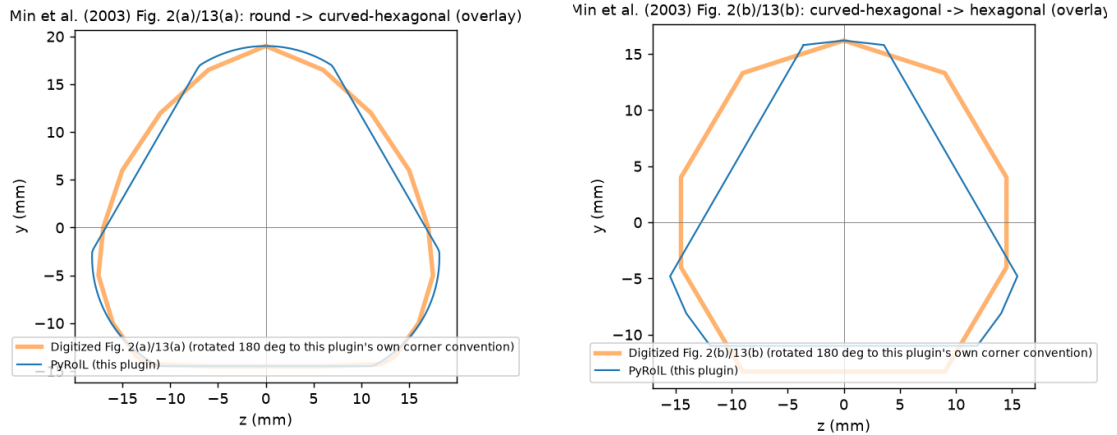
Figure 3: PyRoLL prediction (right) vs. digitized Fig. 9/10 of Byon, Kim, Kim, and Lee [2] (left), 7-pass Kocks mill.

Coordinates were read off 300 dpi page renders of the source PDFs by eye against the

printed axis gridlines; typical reading uncertainty is ± 0.5 –1 mm for the millimeter-scale shape plots and ± 0.01 –0.03 for the dimensionless spread/eccentricity calibration plots (Figs. 7, 9, 12). `tests/test_three_roll_bulge_formulas.py` additionally validates the eccentricity coefficients (2.40, 3.133) directly against the digitized FE calibration data of Figs. 7 and 12, and the closed-form Koval $\rightarrow$ round eccentricity against its defining equations, independent of any PyRoLL geometry construction.

3.1 Overlay comparison

The side-by-side figures above make the overall shape easy to compare but not the small change in slope where a corner’s bulge circle hands off to the plain contour (Section 2.3); Fig. 4 and Fig. 5 instead plot the digitized reference and the PyRoLL prediction directly on top of each other on the same axes. For Min’s data (a right-half trace, mirrored about $z = 0$ to a full boundary and rotated 180° to align the paper’s own drawing orientation with this plugin’s corner-angle convention) the two curves are close enough that the hand-off is not visually distinguishable from the digitized shape’s own reading noise, except for the curved-hexagonal $\rightarrow$ hexagonal pass, where a real, larger size mismatch is visible – a known limitation of the approximate (non-exact) groove parameters used for that test case, not of the construction method itself. For Byon’s data (one sixth-symmetric sector, reconstructed to a full boundary by mirroring and threefold rotation, see `reconstruct_threefold_sector` in `tests/visual.py`) the match is close for passes 2, 3 and 7; pass 1 instead overlays the *fallback* shape against the true digitized bulge, by construction not matching it, as the honest visualization of Section 2.5’s fallback case.



(a) Min et al., round $\rightarrow$ curved-hexagonal.

(b) Min et al., curved-hexagonal $\rightarrow$ hexagonal.

Figure 4: Digitized reference (orange) overlaid on the PyRoLL prediction (blue).

`tests/test_three_roll_end_to_end_spreading.py` complements this with a fully predictive (not pinned-width) demonstration: two-pass sequences for both mill types,

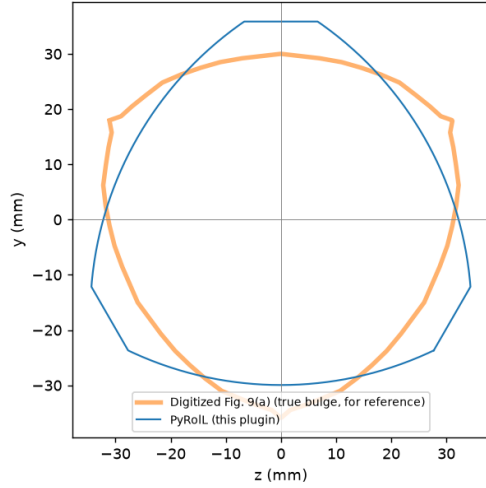
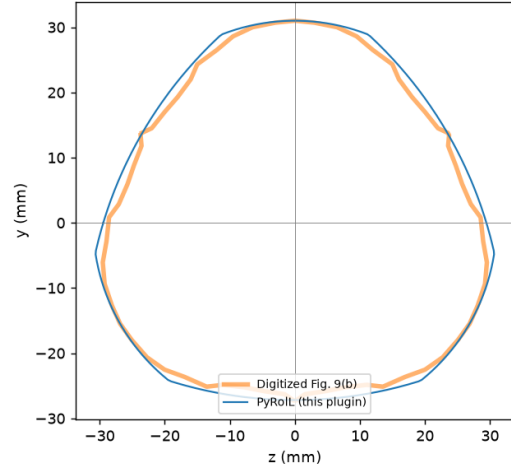
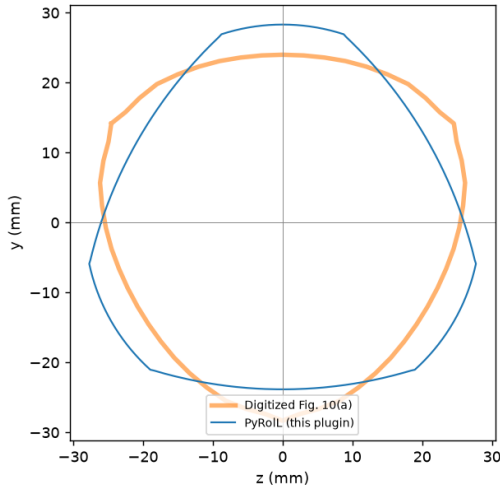
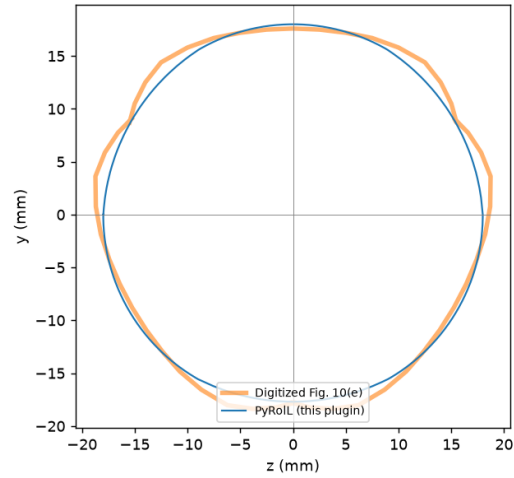
Byon et al. (2017) Fig. 9(a): round $\rightarrow$ Koval (fallback shape vs. digitized)(a) Round $\rightarrow$ Koval (pass 1): fallback shape vs. true bulge.Byon et al. (2017) Fig. 9(b): Koval $\rightarrow$ Koval (overlay)(b) Koval $\rightarrow$ Koval (pass 2).Byon et al. (2017) Fig. 10(a): Koval $\rightarrow$ Koval (overlay)(c) Koval $\rightarrow$ Koval (pass 3).Byon et al. (2017) Fig. 10(e): Koval $\rightarrow$ round (overlay)(d) Koval $\rightarrow$ round (pass 7, final).

Figure 5: Digitized reference (orange, reconstructed from one sixth-symmetric sector) overlaid on the PyRoLL prediction (blue).

with the out-profile width predicted by `pyroll-wusatowski-spreading` fed by `pyroll-lendl-equivalent-method` (Section 1), and this plugin’s bulging model applying on top with no manual intervention, as a real user of the plugin would run it. Since Wusatowski’s is a generic flat-rolling formula rather than Min’s or Byon’s own three-roll-specific spread equation, its predicted width for the Byon-style sequence’s first pass triggers the very fallback described in Section 2.5 (unlike the pinned-width case of Fig. 3(a), which is pinned to the paper’s own, smaller, validated width) – a realistic illustration of the fallback guarding a real spread-prediction chain, not just a hand-picked pinned-width test case.

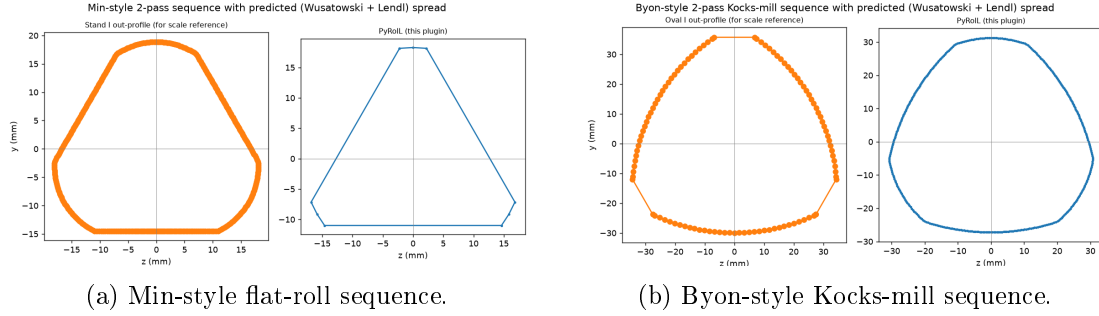


Figure 6: Second-pass out-profile (right) with the width predicted end to end by Lendl + Wusatowski, shown against the first-pass out-profile for scale (left, not a reference shape here).

4 Usage instructions

Importing the plugin registers its post-processor on `ThreeRollPass`; no further setup is needed. Load a spread-prediction plugin and the `pyroll-lendl-equivalent-method` package alongside it (see Section 1) to have the out-profile width predicted rather than requiring it explicitly:

```
from pyroll.core import Profile, Roll, ThreeRollPass, FlatGroove
import pyroll.wusatowski_spreading
import pyroll.lendl_equivalent_method
import pyroll.profile_bulging

in_profile = Profile.round(diameter=35.8e-3, temperature=1323.15, strain=0,
                           material=["C45", "steel"], flow_stress=100e6,
                           length=1, density=7.5e3,
                           specific_heat_capacity=690, thermal_conductivity=23)

roll_pass = ThreeRollPass(
    roll=Roll(groove=FlatGroove(r1=5e-3, usable_width=45e-3, pad_angle=30),
```

```
        nominal_radius=185e-3, rotational_frequency=248 / 60),  
        inscribed_circle_diameter=29e-3,  
    )  
  
out_profile = roll_pass.solve(in_profile)  
print(out_profile.width, out_profile.bulge_radius, out_profile.cross_section.area)
```

Shell command to build this documentation (requires a working LaTeX installation):

```
pyroll input.py solve report
```

References

- [1] J. H. Min et al. “Analytical model for prediction of deformed shape in three-roll rolling process”. In: *Journal of Materials Processing Technology* 140 (2003), pp. 471–477. DOI: 10.1016/S0924-0136(03)00717-9.
- [2] Sang-Min Byon et al. “An approximate model to predict the surface profile of material sections in a 3-roll rolling process”. In: *Journal of Mechanical Science and Technology* 31.7 (2017), pp. 3489–3497. DOI: 10.1007/s12206-017-0638-7.